

Ethan Huber

✉ ehuber@uoguelph.ca 📞 (519) 880 6350 🔗 ethanhuber.ca in ethan-huber 🌐 zushiEdu

Technical Skills

Programming Languages: C, Java, Go, C#, JavaScript, TypeScript, Python, HTML/CSS, VHDL

Tools and Frameworks: Git, .NET, Linux, Angular, Postgres, SQL, REST API, Docker

Design and Planning Tools: OnShape, KiCad, Solidworks, Vivado, Xilinx ISE, AutoCad

Work Experience

Systems Engineering Co-op Analyst

May 2026 – Present

Toyota Motor Manufacturing Canada

Cambridge, Ontario

- Improved and modernized the automated guided vehicle traffic control system's UI by migrating components to the PrimeNG framework, increasing consistency across the application
- Completed a prior co-op's Toyota Business Practice project and led an independent TBP initiative to target regression test inefficiencies that were delaying software releases

Process Control Team Member

May 2025 – May 2026

Toyota Motor Manufacturing Canada

Cambridge, Ontario

- Supported final assembly operations using Toyota Production System principles to maintain workflow efficiency
- Identified 19 potential defects during vehicle shipping preparation (manual installation and label application) demonstrating exceptional attention to detail and commitment to quality

Education

Bachelor of Engineering, Computer Engineering (Co-op)

Sept 2024 – May 2029 (Expected)

University of Guelph; GPA: 3.7

Guelph, Ontario

- Maintaining an 86% average, Dean's Honors List: Winter 2025, Fall 2025, Winter 2026
- Relevant courses: Data Structures and Algorithms, Engineering Systems Analysis, Object Oriented Programming for Engineers, Digital System Design, Software Systems Development and Integration

Key academic project:

- Treasure Runner - 2D Puzzle Game Feb 2026 - Apr 2026
 - Built a 2D game in C with room traversal, pushable objects, and treasure collection exposed to Python via ctypes
 - Designed a graph-based world system with dynamic room loading and player state management

Extracurricular Experience

Electrical Team Lead

Sep 2025 – Present

University of Guelph Robotics Team

Promoted to Lead in May 2026

- Led the electrical sub-team and coordinated with mechanical and software teams to develop, test, and integrate rover electrical systems for the Canadian International Rover Competition
- Designed and routed multiple PCBs in KiCad, optimizing trace sizing, connector placement, and component layout
- Validated board performance through testing and wrote safety documentation to ensure competition compliance

Personal Projects

Modular Agricultural Robot Platform // April 2026 – Present

github.com/zushiEdu/GRP2 🔗

- Designed a sheet metal aluminum rover with custom 3D-printed in-hub gearboxes, modular sensor bumpers, and a swappable payload mount, optimized for low-cost farmer repair and maintenance
- Developed a custom STM32 control board with four MOSFET motor drivers communicating to a Raspberry Pi over UART, with I2C support for modular attachments such as a fertilizer dispenser
- Implemented GPS guided autonomous field navigation with computer vision for efficient agriculture applications

PCB Business Card // Dec 2025 – Jan 2026

github.com/zushiEdu/BusinessCard 🔗

- Designed a PCB using KiCad with an STM32F103 powering a 12x12 LED matrix to simulate Conway's Game of Life
- Programmed C firmware with STM32CubeIDE, and soldered SMD components onto the board

Filedex - Semantic file search engine // Jul 2025 – Sep 2025

github.com/zushiEdu/filedex 🔗

- Developed a semantic file search engine in Go for file indexing and retrieval
- Integrated SQLite for structured metadata storage
- Implemented graph traversal algorithms to identify relationships between tagged files
- Deployed local AI/ML recognition using Ollama to automate metadata tagging generation